

Rishikanth Manimeli

📍 Amherst, MA ✉ rishikanth47@gmail.com ☎ +1-413-313-9396 in rishikanth-0605 🌐 RiKan47
🔗 rikan47.netlify.app

Experience

Samsung R&D

- Senior Software Engineer** | C++, Linux 03/2024 – 01/2025
- Optimized C++ code paths in the 4G eNB uplink MAC stack by localizing frequently accessed data and reducing repeated conditional checks; validated changes with before/after metrics, UE connection tests, and throughput measurements.
 - Contributed to a team effort that improved peak uplink throughput by **14%** for Tier-1 North American carrier programs, with validation at loads including **800 and 1,200 concurrent UEs**.
- Software Engineer** | C++, Python, GTest 06/2022 – 03/2024
- Maintained uplink MAC code through implementation and review of optimization changes, checking protocol behavior and regression results in a latency-sensitive system.
 - Built a configurable Python log-analysis executable that parsed optimization-test output and exported consolidated Excel tables for reviewing MAC uplink results.

Education

University of Massachusetts Amherst *MS in Computer Science* | *GPA: 3.93/4.0* Expected graduation: 12/2026
Coursework: Distributed & Operating Systems, Database Design, Systems for Deep Learning, Machine Learning, Algorithms.
Indian Institute of Technology Patna *BTech in Electrical Engineering* 07/2018 – 05/2022

Projects

- Keystone KV** | Go, TCP/RESP2, Docker 2026
- Built a TCP key-value server with bounded RESP2 parsing, concurrent clients, indexed expiration, checksummed snapshots, and asynchronous replication with sequence validation and reconnect resynchronization.
 - Measured median **51,149 GET/SET operations/sec** with 32 loopback clients on a two-CPU ARM64 host, with snapshots and replication disabled; documented workload, latency distributions, and recovery demos.
- Distributed Stock Trading** | Python, SQLite, Docker, k6 2025 – 2026
- Co-developed a service-based trading application; extended it with atomic SQLite inventory-and-trade commits, durable idempotency keys, conflicting-key detection, and recoverable order-log projections.
 - Verified **2,700 distinct trades across nine k6 runs** with zero HTTP failures and persisted-inventory checks; injected a crash after commit and confirmed retry without a second inventory change.
- Evidence RAG** | Python, FlashRAG, PyTorch, FastAPI 10/2026
- Built a document-QA workbench with BM25, dense and hybrid retrieval, local FlashRAG generation, and a bounded search/read/finish agent with source-quotation validation.
 - Persisted execution traces in SQLite and replayed tool observations without model calls; added per-case EM/F1, evidence coverage and budget reporting, with **37 passing tests** and container checks.

Research Coursework

Hierarchical Agentic Retrieval | Adobe-associated course project 01/2026 – 05/2026

- Contributed literature review and shared dataset/benchmark preparation to a team study of tool-using retrieval for multi-hop question answering.

Technologies

Languages: C++, Python, Go, Java, SQL, TypeScript
Systems & Backend: Linux, TCP/HTTP, RESP2, SQLite, FastAPI, Docker, Git, GitHub Actions, GTest, k6
AI & Web: PyTorch, FlashRAG, Hugging Face Transformers, React

Achievements

Samsung Excellence Award — System Capacity Improvement and Code Optimization (04/2023).
Samsung Professional-level certification — Algorithmic Problem Solving (10/2023).